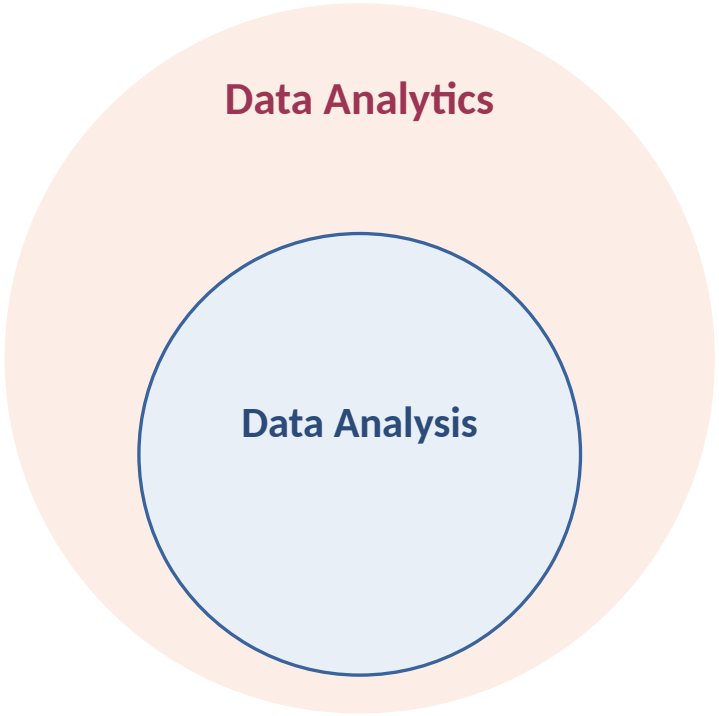
DATA 211 – Introduction to Data Science

EXPLORATORY DATA ANALYSIS

Prof Dr Usman Qamar

Data Analysis and Data Analytics



Analytics is the broader term: it includes analysis.

Data Analysis

- Hands-on data exploration and evaluation
- Looks backwards: a historical view of what has happened
- Not involved in predicting what will happen

Data Analytics

- A broader term that includes data analysis
- The science behind the analysis
- Concerned with the entire methodology

Categorization of Analysis and Analytics

Six types, each answering one question. Rows pair an analysis with an analytics type.

DATA ANALYSIS	DATA ANALYTICS
Descriptive Analysis *What is happening now?*	**Diagnostic Analytics** *Why did something happen?*
Exploratory Analysis *What unknown relationships and patterns exist?*	**Predictive Analytics** *What might happen?*
Mechanistic Analysis *How does a change in one variable change another?*	**Prescriptive Analytics** *What is the best course of action?*

Descriptive and Diagnostic

DATA ANALYSIS

Descriptive Analysis

What is happening now (or has happened so far)?

Analyzes and summarizes the data. We met it in Topic 3 (Basic Statistics):

- Histograms and pie charts to visualize data
- Skew and kurtosis to understand the distribution
- Measures of centrality: mean, median, mode
- Dispersion: range, interquartile range, variance, standard deviation

DATA ANALYTICS

Diagnostic Analytics

Why did something happen?

Done hands-on with a small dataset, it is also called causal analysis.

- Involves at least one cause (usually more) and one effect
- Many techniques are available
- Techniques recognize patterns, detect anomalies and surface ‘unusual’ events
- **Correlation** is one of the most frequently used techniques

Exploratory and Predictive

DATA ANALYSIS

Exploratory Analysis

Find unknown relationships and patterns.

- Typically relies on data visualization
- Plotting the data in different forms gives clues about what we may (or want to) find
- A range of techniques; the most common goal is looking for patterns
- **PCA** and **summary analysis** in this lecture are exploratory

DATA ANALYTICS

Predictive Analytics

What might happen?

It involves four steps:

1. Collect data
2. Clean data
3. Identify patterns
4. Make predictions

Mechanistic and Prescriptive

DATA ANALYSIS

Mechanistic Analysis

How does a change in one variable lead to changes in another?

- Many techniques are available
- **Regression** is one of the most frequently used
- Covered in the Regression unit

DATA ANALYTICS

Prescriptive Analytics

What is the best course of action for a given situation?

It involves three steps:

1. Analyze potential decisions and their interactions
2. Analyze the influences that bear upon these decisions
3. Prescribe an optimal course of action in real time

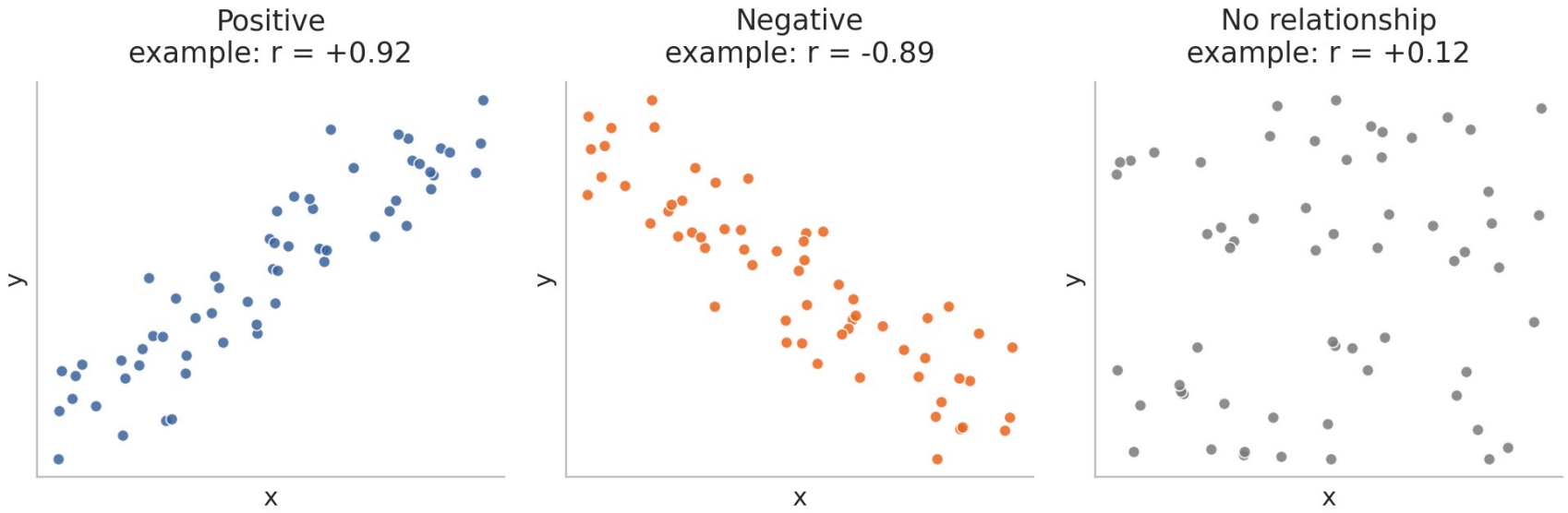
Correlation

Strength

How closely the two variables are related to each other.

Direction

How one variable changes its value as the value of the other variable changes.



Illustrative data. The sign of r gives the direction; its distance from 0 gives the strength.

Pearson's r Correlation

The following formula is used for calculating Pearson's r correlation:

$$r = \frac{N\sum xy - \sum x \sum y}{\sqrt{\left[N\sum x^2 - \left(\sum x\right)^2\right]\left[N\sum y^2 - \left(\sum y\right)^2\right]}}$$

where

Symbol	Meaning
r	Pearson's r correlation coefficient, $-1 \le r \le 1$
N	number of values in each dataset / column
Σxy	sum of the products of the paired scores
Σx, Σy	sum of the x column scores, sum of the y column scores
Σx^2, Σy^2	sum of the squared x scores, sum of the squared y scores

Reading r: Direction and Strength

Direction

Positive: $0 < r \leq 1$

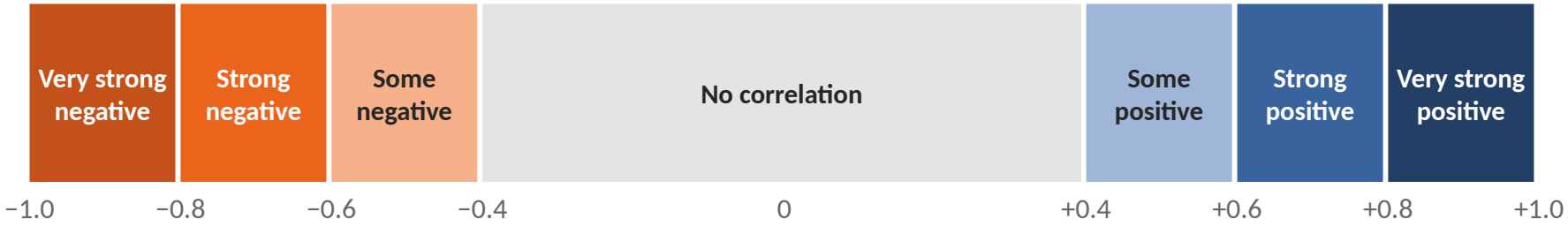
If one increases (decreases), the other increases (decreases).

Negative: $-1 \leq r < 0$

If one increases (decreases), the other decreases (increases).

Strength

The closer r is to 1 or to −1, the stronger the correlation. A guideline (Figure-1):



Example: r = −0.7 is a strong negative correlation; r = 0.9 is a very strong positive one. The sign and the strength are read separately.

Figure-1 after Matthew North, Data Mining for the Masses.

Example: Correlation Coefficient

Question A

A diet company wants to know if there is a relationship between the height and weight of human beings. As a pilot study, they collected data for ten individuals working in the company. The data is given in `EDA-1.csv`.

1. Use Pearson's correlation formula to calculate r between height and weight.
2. Interpret the strength and direction of their relationship.
3. Use Python to verify the above results.

#	Weight	Height
0	64.5	118
1	73.3	143
2	68.8	172
3	65.0	147
4	69.0	146
5	64.5	138
6	66.0	175
7	66.3	134
8	68.8	172
9	64.5	118

EDA-1.csv, all 10 rows

Question A.1: Pearson's r by Hand

Step 1: the column totals (x = Weight, y = Height)

N	Σxy	Σx	Σy	Σx²	Σy²
10	98,335.30	670.70	1,463	45,058.21	218,015

Step 2: substitute into the formula

$$r = \frac{10(98{,}335.3) - (670.7)(1463)}{\sqrt{[10(45{,}058.21) - 670.7^2][10(218{,}015) - 1463^2]}}$$

$$r = \frac{2{,}118.9}{\sqrt{(743.61)(39{,}781)}} = \frac{2{,}118.9}{5{,}438.89} \approx 0.390$$

r ≈ 0.39

The notebook quotes r = 0.389; to three decimals 0.3896 rounds to 0.390.

Question A.2: Interpret the Result

DIRECTION

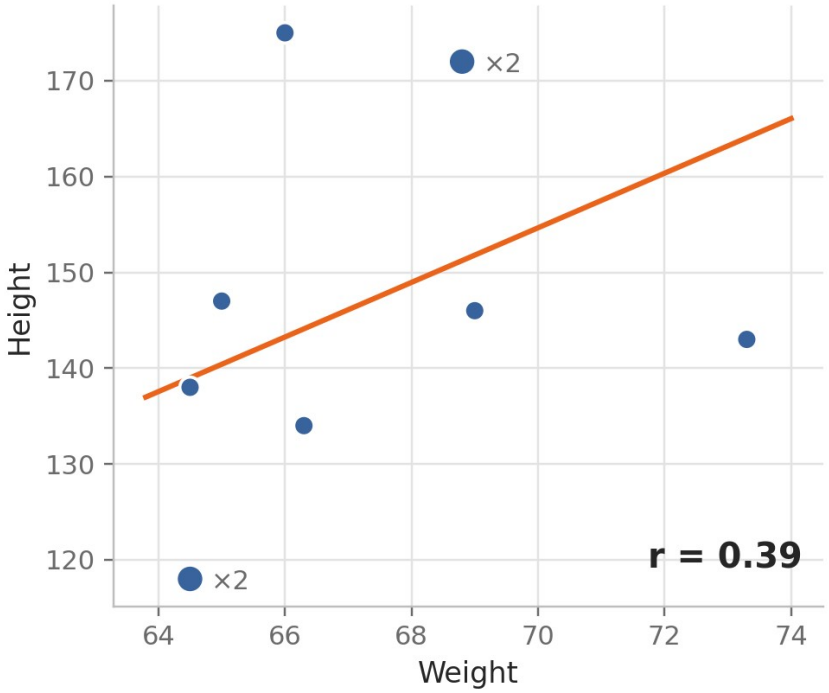
Positive (r > 0)

As Weight increases, Height tends to increase too.

STRENGTH

Some correlation (r ≈ 0.4)

A relationship is present, but the points are widely scattered around the trend.



EDA-1 with a least-squares trend line. ×2 marks a repeated row.

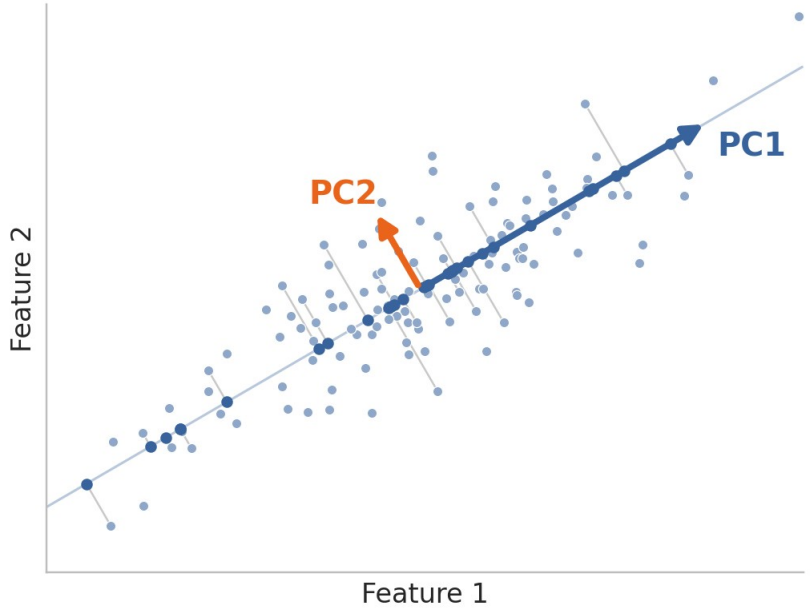
Answer: there is some correlation between the two variables (strength, r ≈ 0.4), and they are positively related (direction, r > 0).

Principal Component Analysis (PCA)

1. A **dimensionality reduction** technique, useful for data visualization and exploration.
2. An **orthogonal projection** of the data onto a lower-dimensional linear space (the principal space) such that the **variance of the projected data is maximized**.

2-D scenario

Each point is projected onto PC1, the direction along which the points spread out the most. PC2 is perpendicular and captures what is left.



Illustrative data: points (light) and their projections onto PC1 (dark).

Intuition Behind PCA

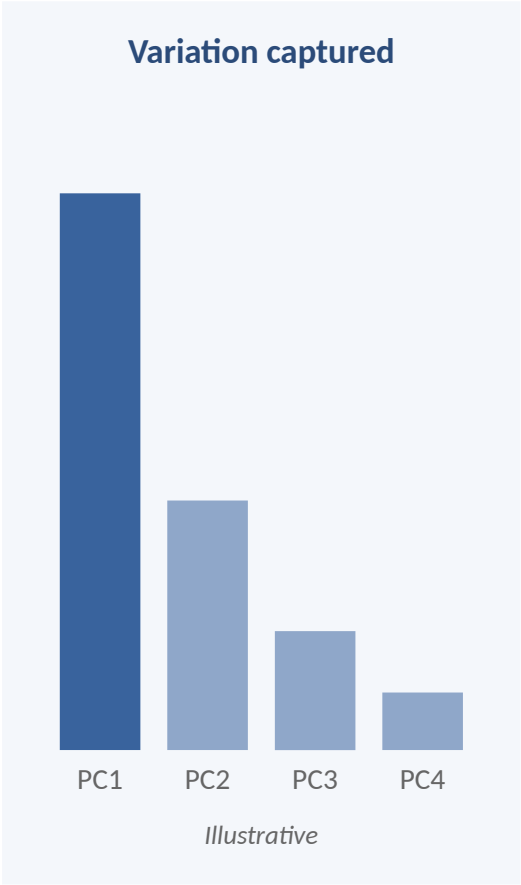
The idea is to find directions that capture maximum variation.

PC1 The first principal component gives the maximum variation, along the principal direction. It is a linear combination of the original columns.

PC2 The second gives the next maximum variation, along a direction perpendicular to the first. It is also a linear combination of the original columns.

PC3 The third gives the next maximum, perpendicular to both the first and second directions.

- All principal components are **ranked** by how much variation they capture.
- Number of principal components = the number of records or the number of columns, **whichever is smaller**.



Obtaining Principal Components

The following steps can be used for obtaining PCs:

1. Find a unit vector that maximizes Var(u). **This is the first PC.**

2. Find the next PC: a unit vector orthogonal to the previously found PCs that maximizes Var(u).

3. Repeat Step 2 until all, or the required number of, PCs are obtained.

Note

Many efficient algorithms obtain PCs. We will use scikit-learn's **sklearn.decomposition** library.

```
from sklearn.decomposition import PCA
```

Case Study 1: PCA

Case Study 1

A car manufacturer wants to identify the relationship between various factors and the *economy* (fuel efficiency) of a vehicle. The data is in `EDA-2.csv`.

1. Read and display the data and data info.
2. Add the first two principal components as columns pca1 and pca2.
3. How much variation does each principal component capture?
4. For PC1: what are the coefficients of the linear combination, and which column has the highest?

Case Study 1 · Step 1: Read the Data

Python

```
#1. Read and display the data & data info.
import numpy as np
import pandas as pd
df = pd.read_csv('data/EDA-2.csv', delimiter = ',')
display(df.head(5))
display(df.info())
```

origin	car_name	model_year	cylinders	displacement	horsepower	weight	acceleration	mpg	economy
american	chevrolet	70	8	307.0	130	3504	12.0	18.0	1
american	buick	70	8	350.0	165	3693	11.5	15.0	0
american	plymouth	70	8	318.0	150	3436	11.0	18.0	1
american	amc	70	8	304.0	150	3433	12.0	16.0	1
american	ford	70	8	302.0	140	3449	10.5	17.0	1

392
rows

10
columns

0
missing values

2
text columns: origin, car_name

From df.info(): 392 non-null entries in every column; dtypes float64 (3), int64 (5), object (2).

Step 2: Add the First Two PCs

```python
num_columns = df.select_dtypes(exclude='object').columns
pca_input_columns = num_columns.drop(['economy'])

from sklearn.decomposition import PCA
pca = PCA(n_components=2)   # get pca model

# get two components
principalComponents = pca.fit_transform(
                          df[pca_input_columns])

df['pca1'] = principalComponents[:,0]
df['pca2'] = principalComponents[:,1]
display(df)
```

Two rules

- Only numerical columns can be used. Encode text columns first, if needed.
- Do not use an OUTPUT column as an input. Here **economy** is the output, so it is dropped.

car_name	cylinders	weight	mpg	economy	pca1	pca2
chevrolet	8	3504	18.0	1	536.47	50.85
buick	8	3693	15.0	0	730.38	79.18
plymouth	8	3436	18.0	1	471.01	75.47
volkswagen	4	2130	44.0	2	−854.71	−6.16
ford	4	2625	28.0	1	−359.54	−35.70

Selected columns of rows 0, 1, 2, 388 and 390; the two new columns are shaded.

Step 3: Variation Captured by Each PC

```python
# variance captured by each PC is in the .explained_variance_ratio_ attribute
for ind,var in enumerate(pca.explained_variance_ratio_):
    print(f'The variance captured by pca {ind+1} is: {var*100: 0.2f}%')
```

99.75%

captured by PC1

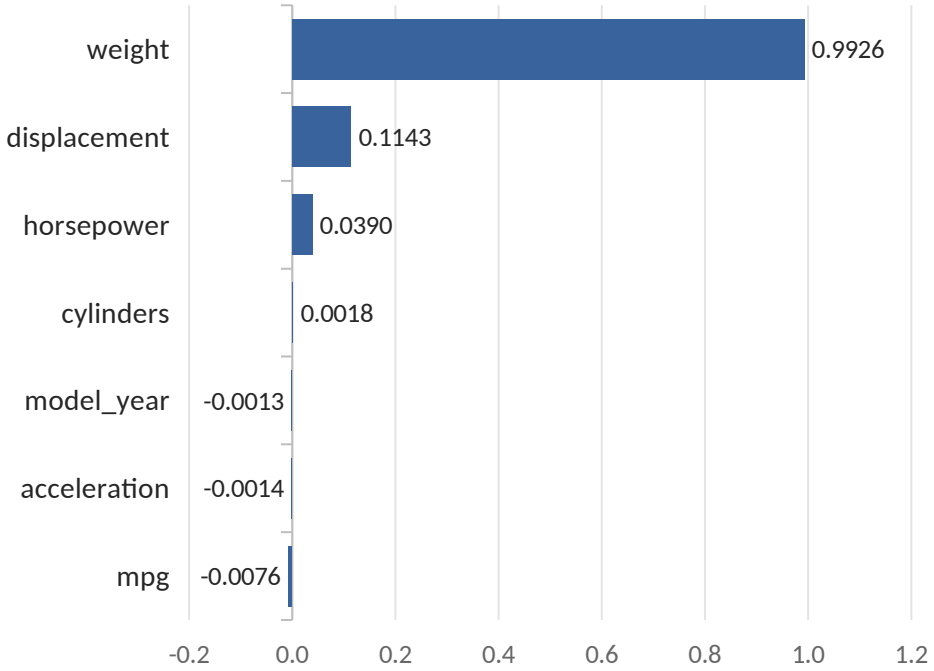
0.21%

captured by PC2

- Together the first two components keep **99.96%** of the variation in the seven input columns.
- Almost everything is in PC1. The next step shows why.

Step 4: Coefficients of PC1

```python
# The coefficients for each PC are in the attribute .components_
pca_1_coefficients = np.round(pca.components_[0], 4)
zipList = list(zip(pca_input_columns, pca_1_coefficients))
ndf = pd.DataFrame.from_records(zipList, columns=['field','coeff'])
display(ndf.sort_values(by=['coeff'], ascending=False))  # consider abs() for negatives
```



Answer: weight has the highest coefficient (0.9926).

Why weight?

PCA chases variance. On unscaled data, PC1 is almost just weight, which is why it captures 99.75%.

End of lecture